

Instructions:

- The exam is open book/note.
- Turn off wifi.
- No AI use of any form is permitted.
- Exam time is 3 hours.
- Write your response to each problem on a separate sheet of paper.
- Show all of your work.

Problem 1. Quadrupole focusing (25 pts)

In accelerators, beam focusing is primarily achieved using alternating focusing and defocusing quadrupole magnets.

- 15 pts: Graphically and mathematically, show why a given quadrupole magnet focuses in one transverse coordinate and defocuses in the other. In your mathematical proof, choose a form of the field and demonstrate the focusing force.
- 10 pts: Consider a doublet consisting of a pair of magnets. In the horizontal (x) direction, the first magnet is focusing and the second magnet is defocusing, as shown in Figure 1. The two magnets are separated by a drift of length L and have equal focal lengths $f_F = f_D = f$.

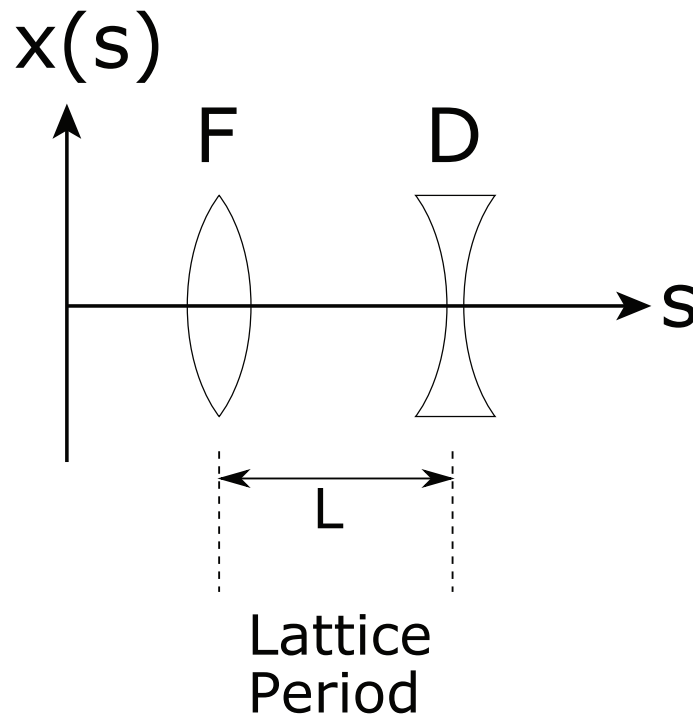


Figure 1: Layout of lenses

Using the transfer matrix description of thin quadrupole lenses, demonstrate that this configuration can achieve net focusing in BOTH the horizontal and vertical directions. (Note: you must show the net focusing effect is achieved in both transverse dimensions for full credit.)

- a) We can show the impact of focusing in the two transverse directions by drawing a field and examining the Lorentz force on a few test particles. In a sketch like Figure 2. below, we can choose a quadrupole field orientation, particle charge, and direction of particle velocity. We can look at particles located on the horizontal and vertical planes.

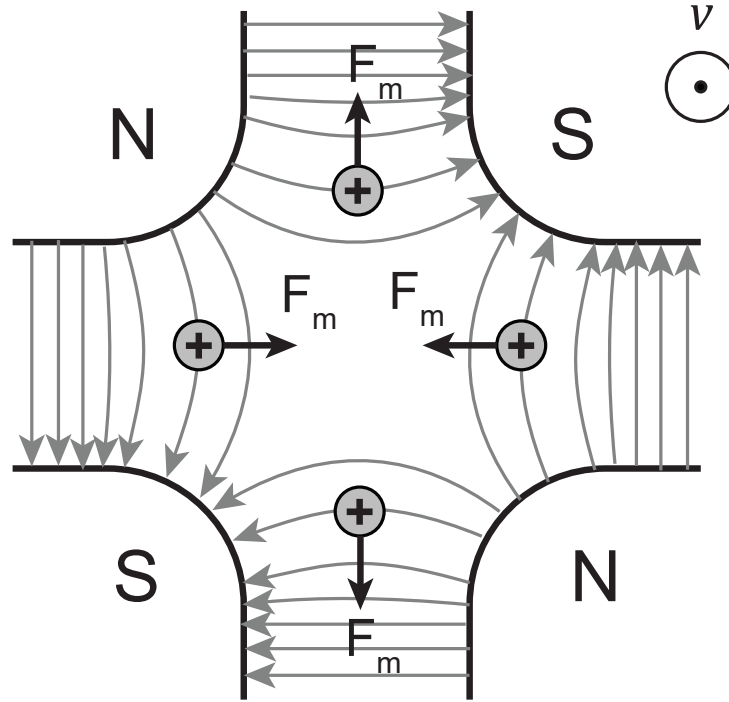


Figure 2: Graphical explanation

The Lorentz force is

$$\vec{F} = q (\vec{v} \times \vec{B})$$

Taking the cross product of the velocity with the field at each of the test particles, we see that in one plane the force is directed towards the center of the magnet (focusing). In the other plane, the force is directed away from the center (defocusing). In the figure shown, this magnet is horizontal focusing and vertical defocusing.

Mathematically, we require that

$$\nabla \times \vec{B} = 0$$

In class, we have shown that this imposes that the gradients of the two coordinates are equal:

$$\frac{\partial B_y}{\partial x} = \frac{\partial B_x}{\partial y}$$

(or z instead of y , depending on choice of notation). Our focusing is related to this gradient. To demonstrate a gradient of this form focuses in one coordinate while defocusing in the other, we can choose a form of the magnetic field which satisfies this relation,

$$B_x(y) = B_1 y, \quad B_y(x) = B_1 x$$

$$\frac{\partial B_x}{\partial y} = \frac{\partial B_y}{\partial x} = B_1$$

Looking first at particles in the x plane ($y = 0$),

$$\vec{B} = B_1(0) \hat{x} + B_1 x \hat{y} = B_1 x \hat{y}$$

Our Lorentz force is

$$\vec{F} = q(v\hat{z} \times B_1 x \hat{y}) = qvB_1(-\hat{x})$$

The force on particles with positive x is in the negative direction towards the center of the magnet. Particles with negative x see a force in the positive $\hat{x}$ direction.

In the y plane ($x = 0$)

$$\vec{F} = q(v\hat{z} \times B_1 y \hat{x}) = qvB_1 y \hat{y}$$

The force on particles with positive y is in the positive $\hat{y}$ direction, away from the center of the magnet. Negative particles are directed downward. This demonstrates this form of a field gradient gives focusing in one transverse coordinate and defocusing in the other.

- b) We need to express this series of elements in matrix form:

$$M = M_{DDriftF}$$

For thin lenses, our transfer matrix is:

$$\mathbf{M} = \begin{bmatrix} 1 & 0 \\ \frac{1}{f} & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & L \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ -\frac{1}{f} & 1 \end{bmatrix}$$

Multiplying, we get

$$\mathbf{M} = \begin{bmatrix} 1 & L \\ \frac{1}{f} & 1 + \frac{L}{f} \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ -\frac{1}{f} & 1 \end{bmatrix}$$

$$\mathbf{M} = \begin{bmatrix} 1 - \frac{L}{f} & L \\ -\frac{L}{f^2} & 1 + \frac{L}{f} \end{bmatrix}$$

We see that we have a negative effective focal length, indicating net focusing:

$$\boxed{\frac{1}{f_{Doublet}} = -\frac{L}{f^2}}$$

In the vertical direction, the lenses instead look like a defocusing lens followed by a focusing lens. This transfer matrix is

$$M = M_{FDriftD}$$

$$\mathbf{M} = \begin{bmatrix} 1 & 0 \\ -\frac{1}{f} & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & L \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ \frac{1}{f} & 1 \end{bmatrix}$$

$$\mathbf{M} = \begin{bmatrix} 1 & L \\ -\frac{1}{f} & 1 - \frac{L}{f} \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ \frac{1}{f} & 1 \end{bmatrix}$$

$$\mathbf{M} = \begin{bmatrix} 1 + \frac{L}{f} & L \\ -\frac{L}{f^2} & 1 - \frac{L}{f} \end{bmatrix}$$

We see that we have the same focusing term,

$$\boxed{\frac{1}{f_{\text{Doublet}}} = -\frac{L}{f^2}}$$

demonstrating that we have net focusing in both directions.

Problem 2. Courant-Snyder Ellipse (25 pts)

In class we derived the single-particle Courant-Snyder invariant:

$$\gamma x^2 + 2\alpha x x' + \beta x'^2 = \epsilon = \text{const.}$$

where:

$$\beta(s) = w^2(s)$$

$$\alpha(s) = -w(s)w'(s)$$

$$\gamma(s) = \frac{1}{w^2(s)} + w'(s)^2 = \frac{1 + \alpha^2(s)}{\beta(s)}$$

Derive the critical values of the ellipse indicated on the figure below:

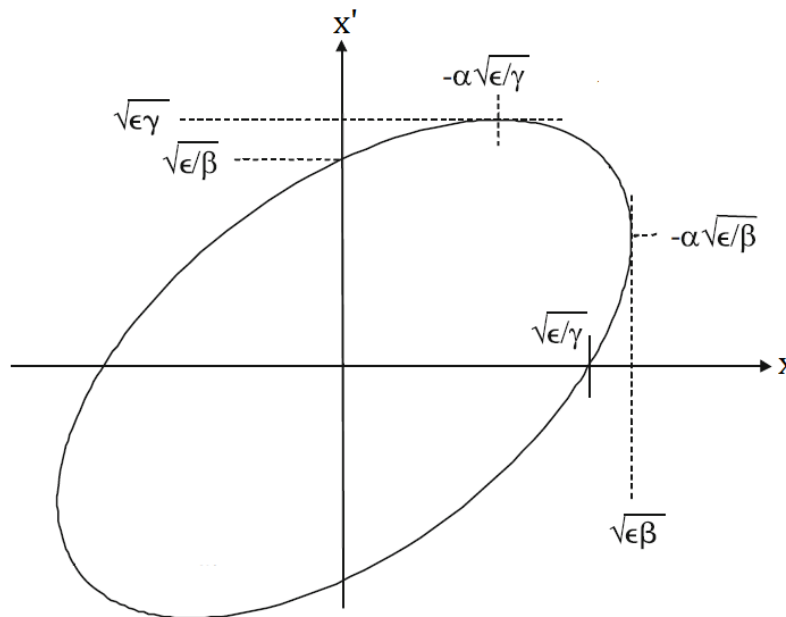


Figure 3: Ellipse

Hint: To avoid messy algebra, take a differential of the constraint equation

$$\gamma x^2 + 2\alpha x x' + \beta x'^2 = \text{const.}$$

as

$$\Rightarrow 2\gamma x dx + 2\alpha x dx' + 2\alpha x' dx + 2\beta x' dx' = 0$$

and use this result to find turning points.

Courant-Snyder Ellipse 25 pts.

The equation of the emittance ellipse is:

$$\gamma x^2 + 2\alpha x x' + \beta x'^2 = \epsilon$$

Where

$$\gamma = \frac{1 + \alpha^2}{\beta}$$

To find the extrema we need to find the points where the first derivative is zero. The full derivative of the ellipse is:

$$2\gamma x dx + 2\alpha x dx' + 2\alpha x' dx + 2\beta x' dx' = 0$$

To find the extrema in x , let $dx = 0$ to get a criteria for x' then plug into ellipse equation:

$$\begin{aligned} \alpha x + \beta x' &= 0 \implies x' = -\alpha x / \beta \\ \implies \beta \gamma x^2 - 2\alpha^2 x^2 + \alpha^2 x^2 &= \beta \epsilon \implies x^2(\beta \gamma - \alpha^2) = \beta \epsilon \\ \beta \gamma - \alpha^2 = 1 &\implies x_{max} = \pm \sqrt{\beta \epsilon}, \quad x'(x_{max}) = -\frac{\alpha x_{max}}{\beta} = \mp \alpha \sqrt{\frac{\epsilon}{\beta}} \end{aligned}$$

To find the extrema in x' , let $dx' = 0$ to get a criteria for x then plug into ellipse equation:

$$\begin{aligned} \gamma x + \alpha x' &= 0 \implies x = -\alpha x' / \gamma \\ \implies \alpha^2 x'^2 - 2\alpha^2 x'^2 + \gamma \beta x'^2 &= \gamma \epsilon \implies x'^2(\beta \gamma - \alpha^2) = \beta \epsilon \\ \beta \gamma - \alpha^2 = 1 &\implies x'_{max} = \pm \sqrt{\gamma \epsilon}, \quad x(x'_{max}) = -\frac{\alpha x'_{max}}{\gamma} = \mp \alpha \sqrt{\frac{\epsilon}{\gamma}} \end{aligned}$$

To find the zeros in x , let $x = 0$.

$$\beta x'^2 = \epsilon \implies x'(x = 0) = \pm \sqrt{\frac{\epsilon}{\beta}}$$

To find the zeros in x' , let $x' = 0$.

$$\gamma x^2 = \epsilon \implies x(x' = 0) = \pm \sqrt{\frac{\epsilon}{\gamma}}$$

Problem 3. Drifting Particles (20 pts)

A proton bunch is injected into a storage ring at a kinetic energy of 10 MeV. The bunch has finite energy spread and bunch length, and no average correlation in time and energy.

- a) 4 pts: What values of synchronous phase could you choose for phase stable motion in the ring?
- b) 4 pts: Sketch the longitudinal phase space in terms of ΔE and z at injection.
- c) 4 pts: Someone forgot to turn on the RF cavity, but at least they didn't forget to turn on the dipoles, so the beam continues to circulate. Sketch how the phase space in b) would evolve in time.
- d) 8 pts: Assume the ring of circumference C contains n_b equally spaced bunches, with the RF off. You may consider the storage ring to be a continuous magnetic field. Derive an expression for the debunching time or the time it takes for the proton bunches to spread out longitudinally until they overlap.

Hint: Look at the phase space you generated in c), expanding the z axis to contain multiple bunches. Then think about how the relative velocity of the particles in the bunch depend on their relative energy, $\Delta v/\Delta E$.

- a) Since the beam is made of protons, γ is very low and the beam energy is well below the transition energy.

$$0 < \phi_s < \pi/2 \quad (1)$$

We have not said if we are accelerating or not, so we don't need to make a distinction of what phase would give acceleration.

- b) Any ellipse or circle in ΔE - z space is valid, as long as it is not tilted at $t=0$.
- c) Due to the finite energy spread, ΔE , each proton travels with a slightly different velocity. So the ellipse rotates/shears, but keeps the same height in the ΔE axis. Here we assume no energy loss to SR since γ is so small.

Below the transition energy, larger energy particles rotate around the ring faster and move ahead and low energy particles will fall behind. So the ellipse should rotate clockwise. (See figure).

- d) From the hint, they should be able to sketch the figure below, and see that the time for the beam to fill the ring is just the time for the particles to spread out by 1/2 the bunch separation in each direction.

The time τ it takes for debunching is equal to the time it takes for the fastest particle to travel half the separation of bunches $\Delta z = C/n_b$ where C is the circumference. We use Δv here because that is the relative speed that a particle moves towards the neighboring bunch.

$$\tau = \frac{\Delta z}{\Delta v} = \frac{C}{2n_b \Delta v}$$

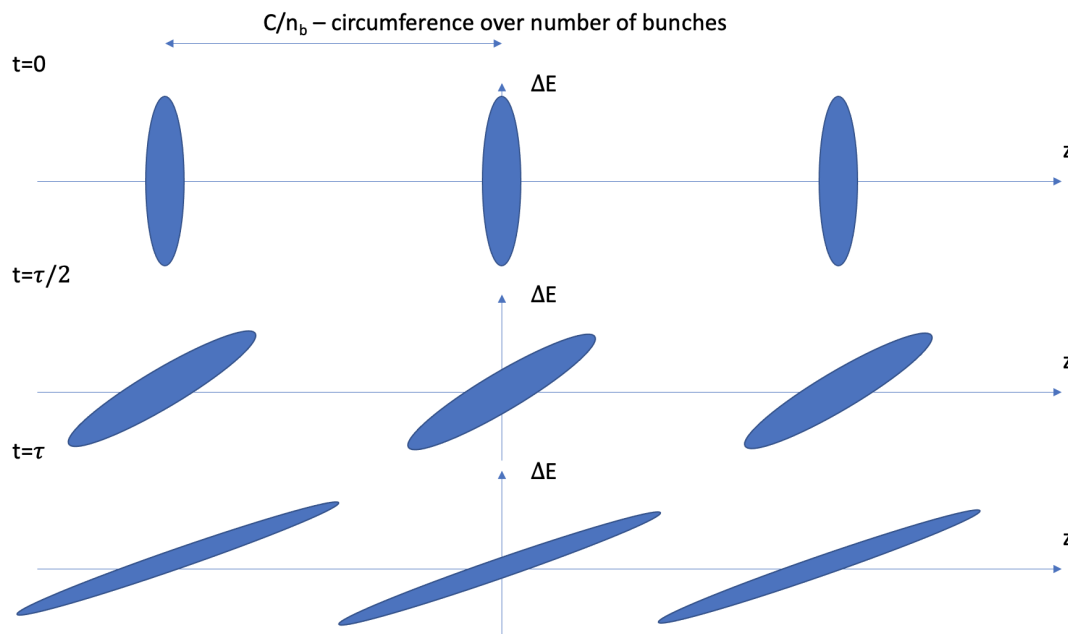
Since the beam is non-relativistic, we can say the kinetic energy is

$$E = \frac{1}{2}mv^2 \implies \Delta E = mv\Delta v$$

$$\frac{\Delta E}{E} = \frac{mv\Delta v}{\frac{1}{2}mv^2} = \frac{2\Delta v}{v} \Rightarrow \frac{1}{\Delta v} = \frac{2E_0}{v_0\Delta E}$$

$$\tau = \frac{C}{2n_b\Delta v} = \frac{C}{n_b} \frac{E_0}{v_0\Delta E}$$

Note - depending on how they define ΔE , you may get variations of constants (1/2, 1/4) in front of this formula, but that's ok.



Problem 4. Off-Momentum Motion (10 pts)

Describe the differences between dispersion and chromaticity in an accelerator lattice. What are the primary sources of each in an accelerator made up of linear optics? What are their primary impacts on the horizontal and vertical motion?

For this problem, you only need to consider the effects that we have discussed so far, based on taking the lowest order in δ .

Looking for (roughly 2 pts each, roughly, other possibilities below)

- Dispersion is quantifying impact from bending on an otherwise ideal trajectory for an off-momentum particle
- Chromaticity arises from differences in focusing for varying particle momentum (chromatic aberration)
- Dipoles/bending introduce dispersion
- Chromaticity comes effective focusing being different

- See both in the horizontal motion, but only chromaticity in the vertical motion

Some other things they may mention that are not necessary, but should count for partial credit if they don't hit all the points above:

- Dispersion can also be introduced by magnet errors, like a misaligned quadrupole
- We express $x(s) = x_\beta(s) + \delta D(s)$
- Betatron oscillations occur around the perturbed closed orbits described by δD (essentially the same as the above, but in words rather than equation form)
- Chromaticity impacts the betatron tune and causes β amplitude modulation
- An accelerator lattice containing quadrupoles has an unavoidable natural chromaticity that must be corrected in storage rings

Problem 5. Synchrotron Radiation (20 pts)

- 7.5 pts: The LHC accelerates proton beams to 7 TeV in a superconducting ring with a 26.7 km circumference. The superconducting bending dipoles provide a 8.3 T field. Calculate:
 - The radiation energy of the 7 TeV protons per turn.
 - The critical energy of photons.
 - The radiation power for a beam current of 800 mA
- 7.5 pts: The LHC tunnel was used for the LEP (Large Electron Project). LEP accelerates the electron beam to ~ 100 GeV, the highest electron energy achieved in a collider. Assuming the dipole layout of LEP is the same as for the LHC, repeat the calculations carried out in part (a) for the electron beam.
- 5 pts: Comment on what changes you would make in building a new e^-e^+ circular collider to reach higher particle energy.

A list of potentially useful constants is provided below.

c	3×10^8 m/s	
m_p	938 MeV/ c^2	1.67×10^{-24} g
m_e	511 keV/ c^2	9.11×10^{-28} g
e	1.60×10^{-19} C	
$\hbar/2\pi$	6.58×10^{-16} eV·s	1.06×10^{-34} J·s
proton r_c	1.53×10^{-18} m	
electron r_c	2.82×10^{-15} m	

Synchrotron Radiation 20 pts.

a) 7.5 pts:

Numbers summarized in Table 1. The radiation energy per turn is given by:

$$U = C_\gamma E^4 / \rho = \frac{4\pi}{3} r_c m c^2 \beta^3 \frac{\gamma^4}{\rho}$$

But C_γ was given in terms of electrons. So they can either use the factor C_γ for electrons, and then use the ratio of masses to convert, or start from the original formula and recalculate from the proton physical constants. Note that ρ **must** be the bend radius, not the average ring radius.

$$\gamma = \frac{E}{mc^2} - 1 = 7462$$

$$\rho = \frac{cp}{0.3B} = 2811 \text{ m}$$

$$U_p = \frac{U_e}{1.14 \times 10^{13}} = \frac{1}{1.14 \times 10^{13}} \times (88.5 \text{ keV}) (1.14 \times 10^{13}) \frac{7000^4}{2810} = 6.6 \text{ keV}$$

or, using the proton classical radius:

$$U_p = \frac{4\pi}{3} (1.530^{-16} \text{ m}) (938 \text{ MeV}) (1)^3 \frac{(7462)^4}{2811 \text{ m}} = 6.6 \text{ keV}$$

The critical photon energy is:

$$\omega_c = \frac{3c}{2} \frac{\gamma^3}{\rho} = \frac{3}{2} c \frac{7462^3}{2811} = 66.5 \times 10^{15} \text{ s}^{-1}$$

$$E_c = \frac{3\hbar\gamma^3 c}{2\rho} = \hbar\omega_c = 44 \text{ eV}$$

The average radiated power is

$$P = UI = (6.6 \text{ keV})(0.8 \text{ A}) = 5.28 \text{ kW}$$

where U is the energy and I is the average beam current.

b) 7.5 pts: Similar for electrons, the results are summarized in the table. ρ is the same.

$$\gamma = \frac{E}{mc^2} - 1 = 1.957 \times 10^5$$

$$U_e = 88.5 \frac{100^4}{2811} = 3.1 \text{ GeV}$$

$$E_c = \frac{3\hbar\gamma^3 c}{2\rho} = \hbar\omega_c = 0.79 \text{ MeV}$$

$$P = UI = (3.1 \text{ GeV})(0.8 \text{ A}) = 2.5 \text{ GW}$$

c) 5 pts: Larger radius to reduce power - FCC-ee.

Table 1: Radiation of LHC and LEP

	LHC	LEP
Energy	7 TeV	100 GeV
γ	7462	1.957×10^5
Dipole radius	2811 m	
Radiation energy	6.6 KeV	3.1 GeV
Critical photon energy	44 eV	0.79 MeV
Radiation power @0.8A	5.3 KW	2.5 GW